

14. Volume production and quality analysis testing

Substrate preparation

The manufacturing environment must be completely free of organic, acidic, metallic and alkali substances. Even an impurity of order 1 in 10^6 ppm (parts per million) can drastically affect the performance of the device.

To prepare wafers, polycrystalline silicon containing elements that can modify its conductivity is melted. Then the molten silicon is pulled up by a crystal seed to grow silicon crystals, also called 'ingots,' which are subsequently sliced to get wafers.

The wafers so obtained are coarse and need polishing to make the surface shiny and smooth. Polishing is usually done on one side of the wafer and further processing is done on the smooth surface so obtained. This entire process is known as the Chechroloski (CZ) crystal-growth process. The resulting substrate is shown in Fig. 1.

SiO₂ insulating layer preparation

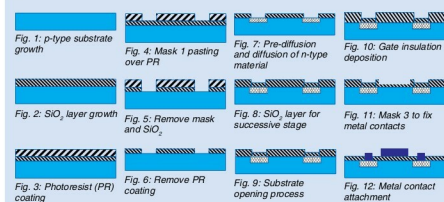
The initial silicon wafers are then subjected to a temperature as high as 1100°C in the presence of oxygen to get the SiO₂ layer deposited over the substrate as shown in Fig. 2:



This SiO₂ layer acts as an electrical insulator as well as mechanical platform for circuits that are built on subsequent layers. As the complexity of the circuits increases, different functional layers will be stacked together, each separated by this SiO₂ layer. Usually, the SiO₂ layer is formed by chemical vapour deposition or physical vapour deposition technique.

Photolithographic process-1

Application of photo-resist. An ultraviolet light (UV) dissolvable photoresist is coated over the wafer on the entire surface as shown in Fig. 3. The coating becomes weak exactly where the UV rays pass through and can be



removed by dissolving in a solvent. At other places, the photo-resist is left intact.

Application of initial mask. A sheet of UV mask resembling the layout of the proposed layer prepared by computer-aided design (CAD) tools is pasted over the photoresist as shown in Fig. 4. This mask has openings, viz, windows, where UV rays have to pass through and fall on the photoresist beneath it. UV rays cannot pass through the mask in other protected areas.

The mask is actually a transparent high-purity glass with the layout of the circuit (created using CAD software) etched onto one side, while the other side is transparent to UV rays. When a UV ray is passed for about 10 minutes, the mask creates an impression on the photoresist through the opened areas which corresponds to the circuit's layout. The photoresist on the areas where UV rays fall becomes acidic and weak and can be removed by dissolving in an alkaline solvent for about 30 minutes. Now the image from the mask has been temporarily transferred to the resist.

Oxide etching. The resist acts as a mask for etching the underlying SiO₂ oxide in an acid bath, usually hydrofluoric (HF) acid. The HF acid is neutral on pure silicon and corrosive on silicon dioxide layer. Hence the SiO₂ layer will be removed by the HF acid where the layer is visible through the openings and remain intact where it is hidden. This is shown in Fig. 5. Subsequently, the UV mask is also peeled off

by some mechanical means.

Now the visible SiO₂ area represents the layout. These SiO₂ layers are removed by a process called de-oxidation using soft acids like HCl. This leaves the substrate prepared for the first layer of impurity deposition. At other places, SiO₂ is left intact and will act as an insulating layer between the substrate and other layers or between successive layers. The resulting material is shown in Fig. 6.

Impurity predeposition

Now the trivalent impurity is transferred from a dopant source (boron nitride, liquid boron tribromide, diborane gas or phosphorus oxide ceramic disks) to the wafer in a precisely controlled temperature and timing arrangement through a process called predeposition or prediffusion. Prediffusion ensures that precise quantity of dopants gets transferred from the dopant source to the wafer. Fig. 7 shows prediffusion and diffusion processes.

Impurity deposition and de-oxidation

Now the dopant is allowed to diffuse uniformly, by a centrifugal rotating arrangement, over the predetermined substrate to form source and drain terminals of the MOSFET at a temperature of about 1100°C. External contacts are inserted and brought out for the drain and source at each successive layer. In order to create successive layers, on this another layer of SiO₂ layer is created by the oxidation